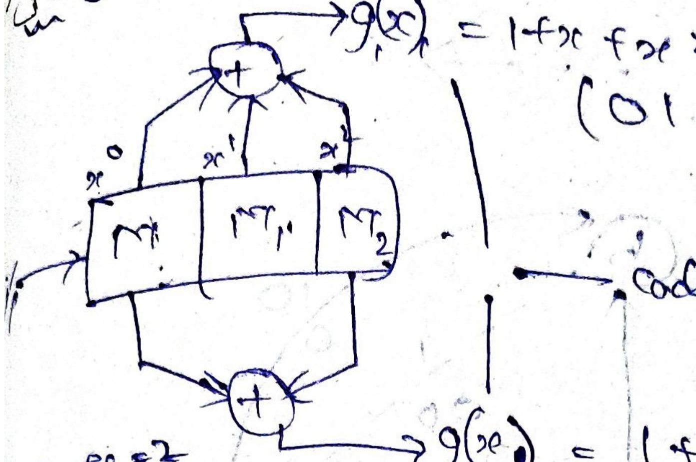
State Diagram, Trellis Diagram:



$g_1(x) = 1+x+x^2$ (01101001)

code word.

$g_2(x) = 1+x^2$ (01100110)

shift reg = 2

$2^n = 2^2 = 4$

M	M_1	M_2	$M \oplus M_1 \oplus M_2$ $g_1(x)$	$M \oplus M_2$ $g_2(x)$
0	0	0	0	0
1	0	0	1	1
0	0	1	1	1
1	0	1	0	0
0	1	0	1	0
1	1	0	0	1
0	1	1	0	1
1	1	1	1	0

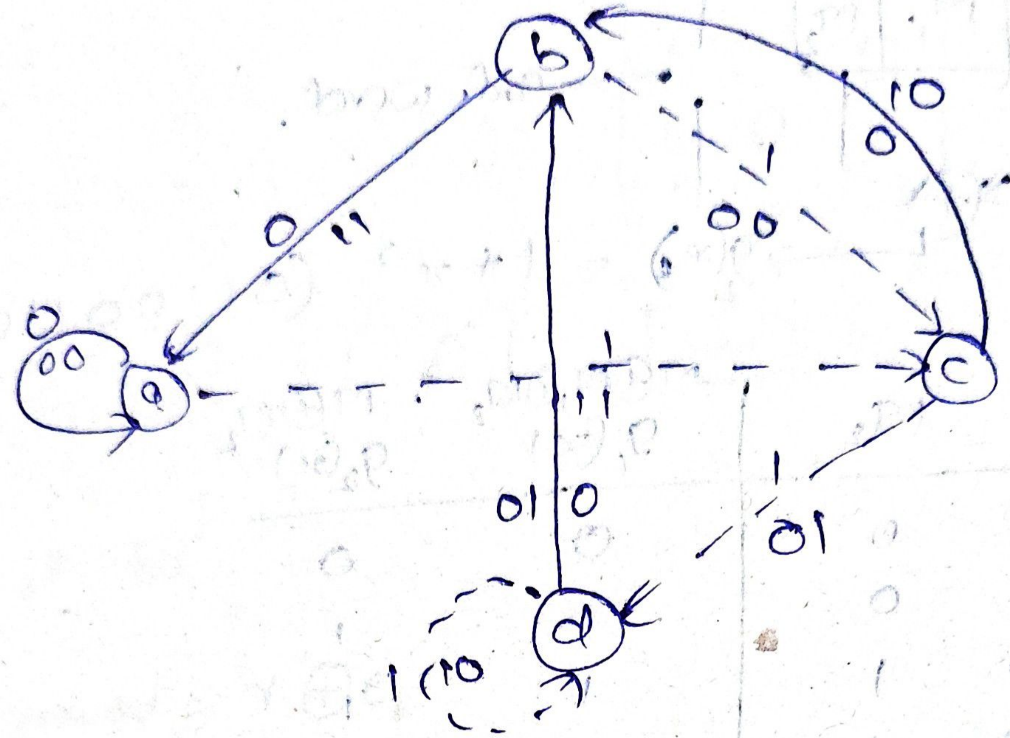
$\therefore$ code word = {00, 11, 11, 00, 10, 01, 01, 10}

$\{x_1^0 x_2^0, x_1^1 x_2^1, x_1^2 x_2^2, \ldots\ldots x_1^6 x_2^6, x_1^7 x_2^7\}$

00 – a
01 – b
10 – c
11 – d

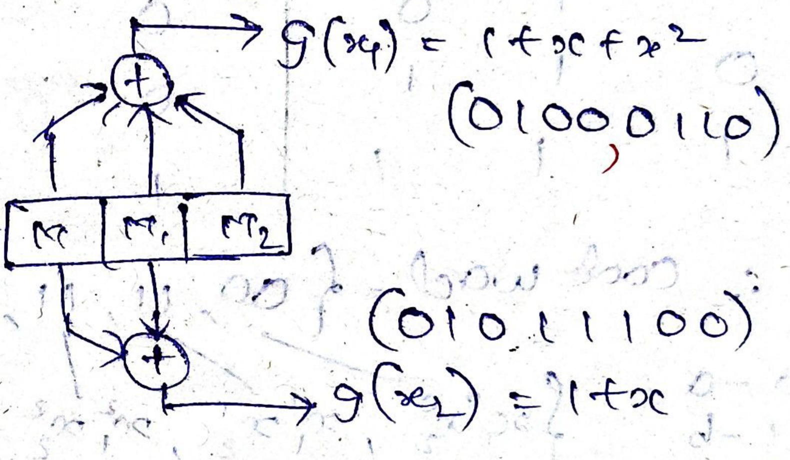
M	M_1	M_2	$g_1(x)$	$g_2(x)$	Present state $M_1 M_2$	Next state $M M_1$	M
0	0	0	0	0	a	a	0
1	0	0	1	1	a	c	1
0	0	1	1	1	b	a	0
1	0	1	0	0	b	c	1
0	1	0	1	0	c	b	0
1	1	0	0	1	c	d	1
0	1	1	0	1	d	b	0
1	1	1	1	0	d	d	1
					d		

i/p
i/p is 1 —teen



Example 1: let input, m = 1101

00 - a
01 - b
10 - c
11 - d

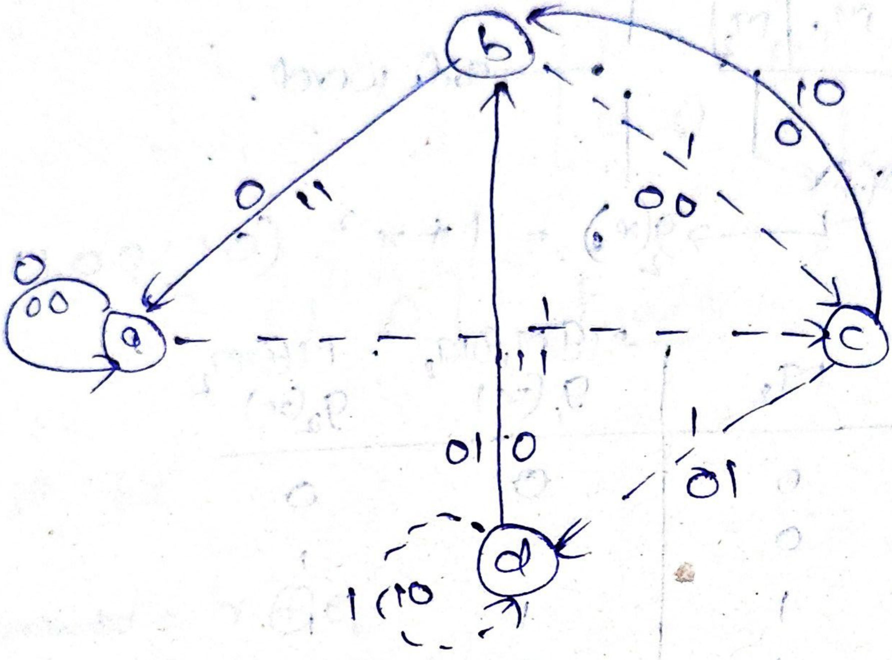


m	m_1	m_2	$g(x_1)$	$g(x_2)$	P.S $m_1 m_2$	N.S $m m_1$	m
0	0	0	0	0	a	a	0
1	0	0	1	1	a	c	1
1	1	0	0	0	c	d	1
0	1	1	0	1	d	b	0
1	0	1	0	1	b	c	1
0	1	0	1	1	c	b	0
0	0	1	1	0	b	a	0
0	0	0	0	0	a	a	0

State diagram:

i/p is 0 then ⟶

i/p is 1 then - - - ->



Example 1:

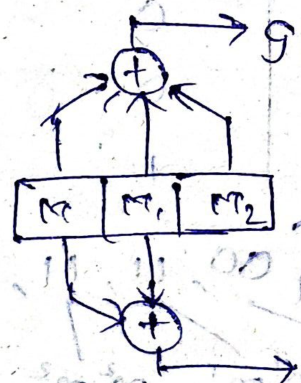
let input, m = 1101

00 - a

01 - b

10 - c

11 - d



$g(x_1) = 1 + x + x^2$ (0100,0110)

(010,11100)

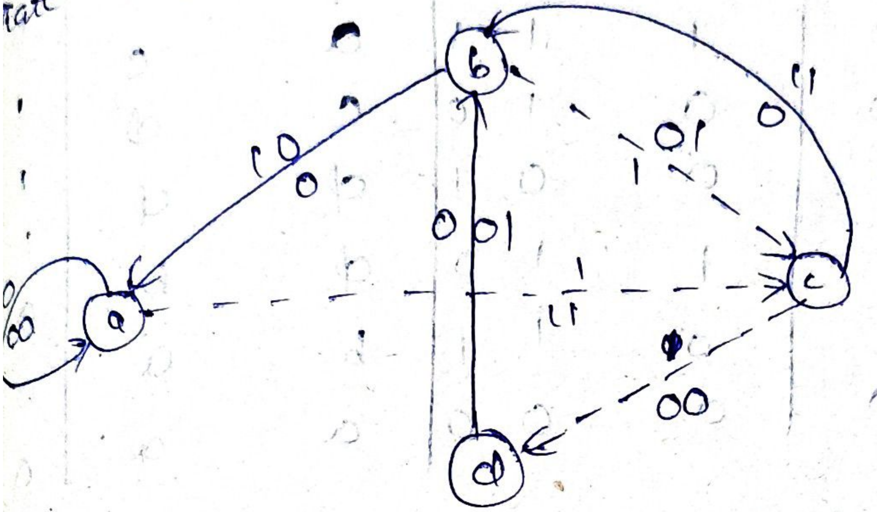
$g(x_2) = 1 + x$

m	m_1	m_2	$g(x_1)$	$g(x_2)$	P.S $m_1 m_2$	N.S $m m_1$	m
0	0	0	0	0	a	a	0
1	0	0	1	1	a	c	1
1	1	0	0	0	c	d	1
0	1	1	0	1	d	b	0
1	0	1	0	1	b	c	1
0	1	0	1	1	c	b	0
0	0	1	1	0	b	a	0
0	0	0	0	0	a	a	0

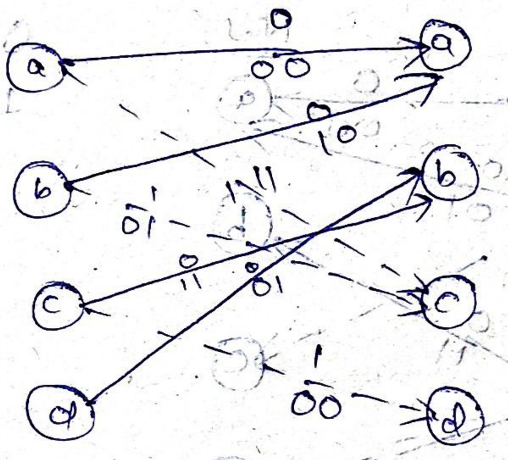
∴ code word = {00, 11, 00, 01, 01, 11, 10, 00}

State diagram,

I/P - 0 ⇒ ⟶
1 ⇒ - - - →

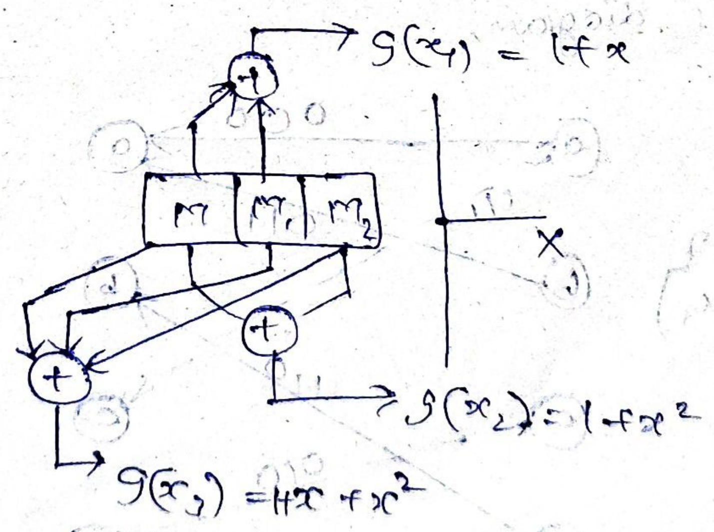


Trellis Diagram,



Example - 2: Let input, m = 1110

00 - a
01 - b
10 - c
11 - d

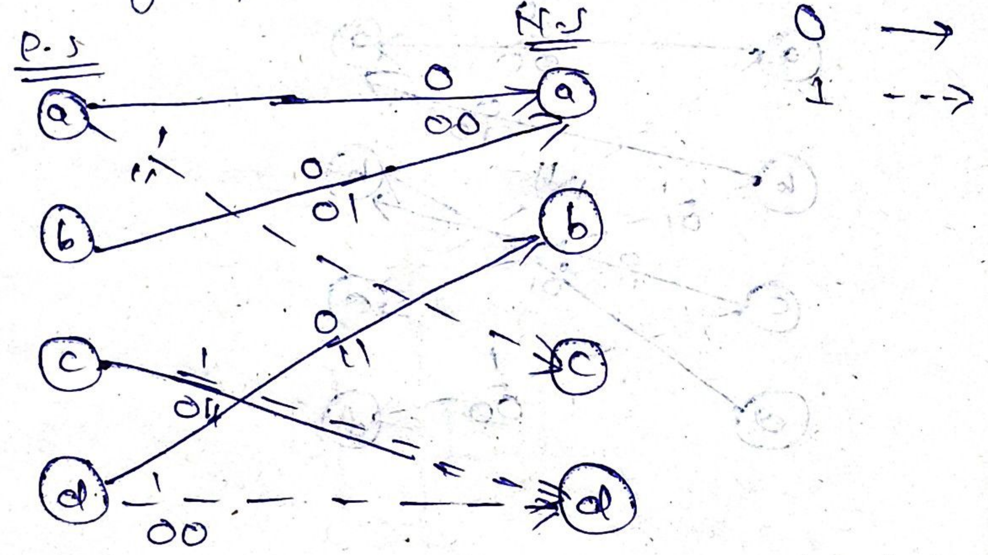


1 M	x M_1	x^2 M_2	$1+x$ $g(x_1)$	$1+x^2$ $g(x_2)$	$g(x_3)$	P.S M_1M_2	N.S MM_1	M
0	0	0	0	0	0	a	a	0
1	0	0	1	1	1	a	c	1
1	1	0	0	1	0	c	d	1
1	1	1	0	0	1	d	d	1
0	1	1	1	1	0	d	b	0
0	0	1	01	1	1	b	a	0
0	0	0	0	0	0	a	a	0

∴ Code word = {00, 11, 01, 00, 11, 01, 00}

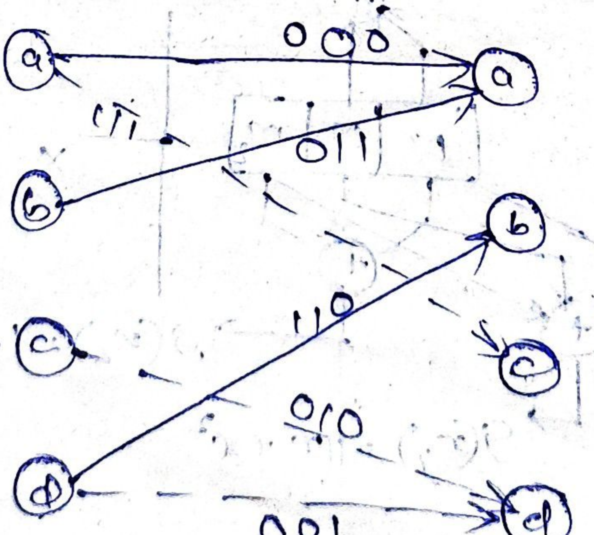
Trelli's diagram,

{For x_1, x_2}



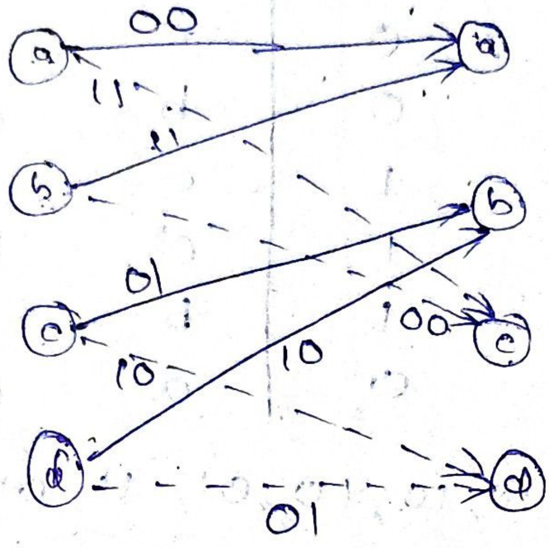
Trelli's diagram,

{For x_1, x_2, x_3}

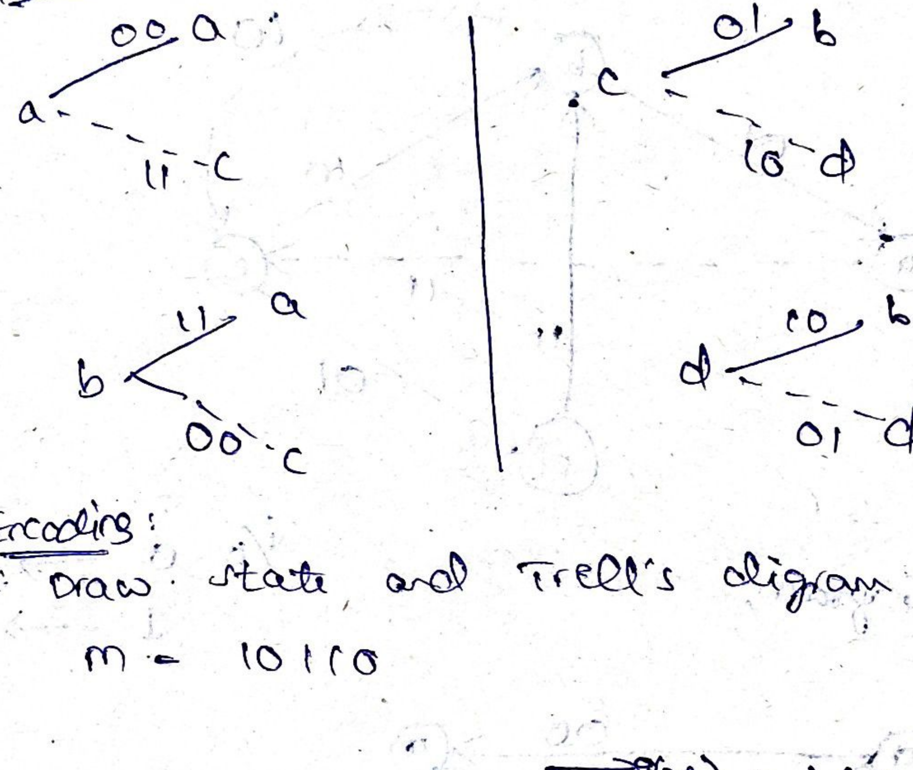


∴ codeword = {000, 111, 010, 001, 110, 011, 000}

Viterbi Algorithm:



General case:

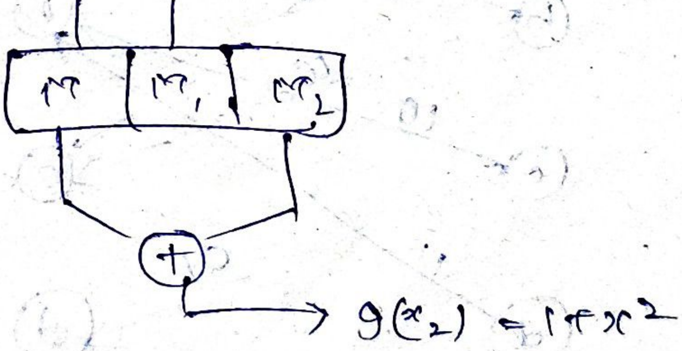


Encoding:

Ex: Draw state and Trelli's diagram for below.

m = 10110

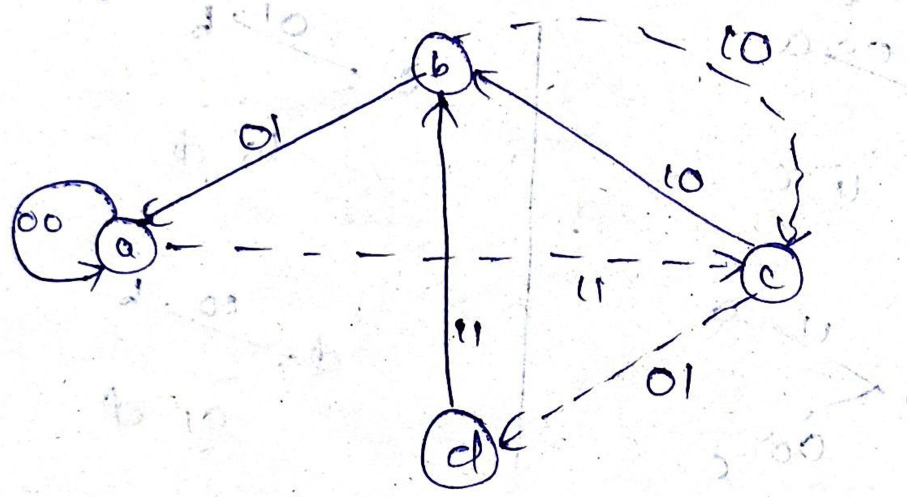
a - 00
b - 01
c - 10
d - 11



1 M	x M_1	x^2 M_2	$1+x$ $g(x_1)$	$1+x^2$ $g(x_2)$	P.S M_1M_2	N.S MM_1	M
0	0	0	0	0	a	a	0
1	0	0	1	1	a	c	1
0	1	0	1	0	c	b	0
1	0	1	1	0	b	c	1
1	1	0	0	1	c	d	1
0	1	1	1	1	d	b	0
0	0	1	0	1	b	a	0
0	0	0	0	0	a	a	0

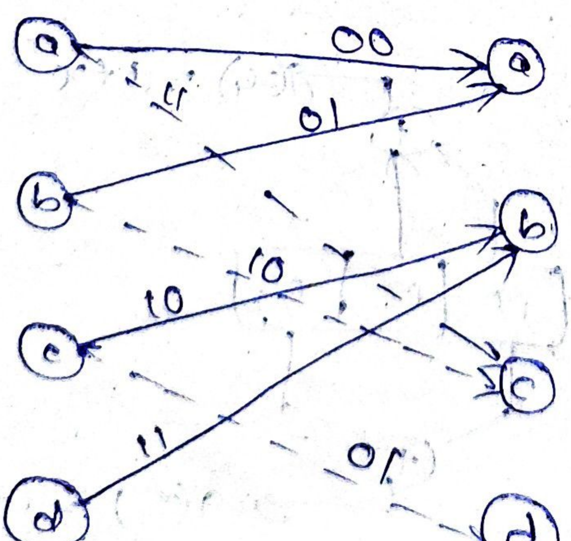
code word = {00, 11, 10, 10, 01, 11, 01, 00}

State diagram,



Trellis diagram,

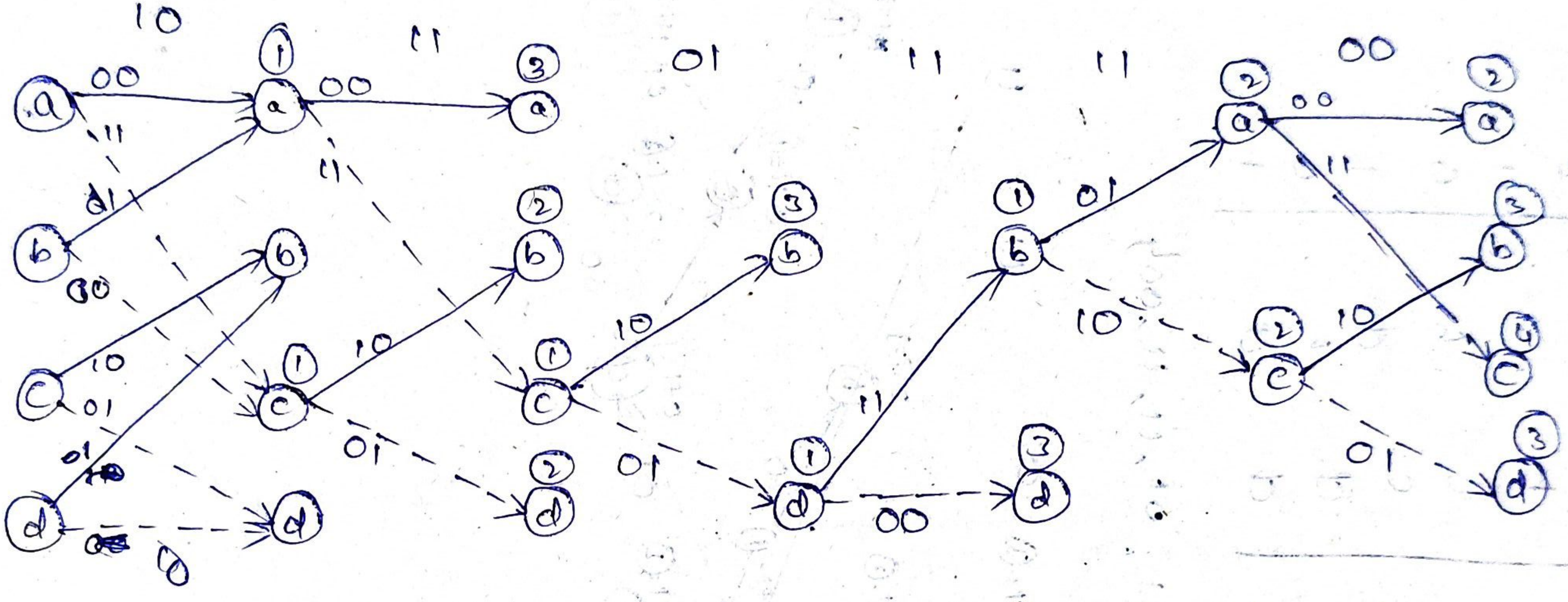
i/p 0 ⟶
1 - - ->



~~for fixed~~ ... (~~if input not given~~)} Viterbi Alg:

m	m_1	m_2	x_1	x_2	m_1m_2	$m\,m_1$	m
0	0	0	0	0	a	a	0
1	0	0	1	1	a	c	1
0	0	1	0	1	b	a	0
1	0	1	1	0	b	c	1
0	1	0	1	0	c	b	0
1	1	0	0	1	c	d	1
0	1	1	1	1	d	b	0
1	1	1	0	0	d	d	1

Consider, {10, 11, 01, 11, 11, 00}



Survival - path : $a \to a \to b \to d \to c \to a \to a$

0 0 0 1 1 0

{10, 11, 01, 11, 11, 00}

00, 01 11 01 11 . 00

So, there is error at 1, 3, 5, 7 position.

∴ correct codeword is {00, 01, 11, 01, 11, 00}